

GroupA_Ticket1ProductVision_CS3203Fall2026

Course: CS 3203 – Software Engineering (Fall 2026)

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Deliverable: Ticket 1 – Product Vision

Product Concept: LinkStream (GoLinks for Developer Teams)

Team Name / Group ID: A

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Team Members & Contributions

Member Name	Role & Core Contribution
Sithabiso Hlanze	Sprint master
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1. Executive Summary & Product Overview

Modern software engineering teams rely on dozens of disparate cloud services daily, including GitHub, Jira, Notion, Figma, Datadog, AWS Console, and Postman. As codebases and team rosters grow, knowledge fragmentation worsens: critical dashboards, specification documents, and staging environments get buried in chat histories or obsolete bookmarks.

LinkStream is a lightweight, cloud-native internal shortcut-redirector and knowledge-navigation service. It enables software teams to turn long, complex URLs into intuitive, memorable shortcuts (e.g., [go/sprint](#), [go/api-docs](#), [go/deploy-staging](#)). By unifying key-value redirection with access control and real-time usage analytics, LinkStream eliminates link-hunting friction and accelerates developer onboarding.

2. Product Brainstorming & Essential Domain Questions

To develop a domain-aligned product, our team brainstormed three fundamental questions addressing developer productivity and enterprise knowledge retrieval:

1. Information Retrieval & Tool Fragmentation:

- *Question:* How do distributed engineering teams currently store, discover, and share internal URLs across distinct cloud platforms without creating single-point-of-failure communication bottlenecks (e.g., constantly asking teammates in Slack)?
- *Insight:* Teams currently rely on unorganized browser bookmarks, static Notion lists, or ad-hoc Slack messages—none of which are universally synchronized, searchable, or trackable.

2. Namespace Collisions & Multi-Tenancy:

- *Question:* How should the system handle duplicate link shortcut names when multiple sub-teams (e.g., Frontend, Backend, QA) want intuitive names like [go/standup](#) or [go/logs](#)?
- *Insight:* The system requires a scoped namespace architecture—offering both global company links ([go/handbook](#)) and team-scoped links ([go/eng/standup](#) or [go/qa/board](#)) with role-based access control (RBAC).

3. Link Health & Observability:

- *Question:* How do we prevent link rot (stale, 404ing destination targets) and verify which resources provide genuine value to the team?

- *Insight:* The platform must incorporate automated HTTP health-check background jobs and a telemetry analytics engine to track click counts, last-accessed timestamps, and unreferenced-link warnings.

3. The 3 Ws Analysis

Following Sommerville Chapter 1 principles, the system is analyzed across three core dimensions:

Dimension	Description
WHAT It Does	• Resolves custom HTTP/DNS redirects (go/<key>). • Searchable, tag-indexed internal dashboard. • Real-time click telemetry and broken-link QA.
WHO It Serves	• Software Developers & DevOps Engineers. • Engineering Managers & Tech Leads. • Newly onboarded engineers and cross-team staff.
WHY It Is Needed	• Recovers lost developer productivity. • Eliminates repetitive link-request messaging. • Speeds up developer time-to-first-commit.

3.1 WHAT the Product Does

- **Instantaneous URL Redirection:** Maps human-readable keys (go/repo) to canonical destinations via a low-latency routing proxy.
- **Central Link Catalog & Search:** Offers a responsive web dashboard with full-text search, categorical tags (#devops, #design, #sprint), and owner metadata.
- **Link Observability & Telemetry:** Monitors redirect frequencies, surfaces unused shortcuts, and alerts owners when destination URLs return 4xx/5xx HTTP errors.

3.2 WHO the Target Users Are

- **Primary Users (Software Engineers):** Developers jumping between bug trackers, pull requests, staging instances, and API documentation daily.
- **Secondary Users (Engineering Managers / Tech Leads):** Leaders distributing single sources of truth for sprint boards, architecture decision records (ADRs), and sprint retrospectives.
- **Tertiary Users (New Hires / Cross-Functional Partners):** Team members needing direct access to essential tools on Day 1 without manual link curation.

3.3 WHY It Is Needed & Value Proposition

- **Reduces Context Switching:** Developers spend an estimated 15–20 minutes daily searching for project artifacts. LinkStream centralizes access in a sub-second lookup.
- **Eliminates Communication Friction:** Eliminates repetitive *"Where is the link to the dashboard?"* inquiries across communication channels.
- **Streamlined Onboarding:** New team members only need to remember [go/welcome](#) or [go/setup](#) to find every prerequisite repository and guide.

4. Structured Product Vision Statement

- **FOR** agile software development teams and engineering organizations
- **WHO** struggle with fragmented cloud toolchains, lost documentation, and high onboarding friction
- **THE** LinkStream platform
- **IS A** centralized, cloud-native internal shortcut redirector and knowledge navigation service
- **THAT** provides instantaneous [go/](#) short-link routing, tag-based resource discovery, and real-time usage analytics
- **UNLIKE** static browser bookmarks, disjointed shared documents, or ad-hoc Slack link dumps
- **OUR PRODUCT** offers automated broken-link detection, scoped team namespaces, and effortless link creation that keeps technical documentation organized and immediately

accessible.

5. Low-Fidelity Prototype & System Workflow

Shortcut	Target Destination	Tags	Clicks	Actions
go/sprint	https://jira.company.com/secure/Rapid..	#agile	1,420	[Edit]
go/api-docs	https://api.company.com/v1/swagger-ui	#backend	890	[Edit]
go/figma	https://figma.com/files/project-alpha	#design	610	[Edit]
go/deploy	https://argocd.internal.cloud/apps	#devops	430	[Edit]

Core User Journey

1. **Creation:** A developer opens the web portal, enters a short key ([go/design](#)), pastes the Figma URL, assigns tags, and clicks **Save**.
2. **Resolution:** Any engineer entering [http://go/design](#) into their browser URL bar is immediately redirected by the routing proxy to the Figma workspace.
3. **Discovery:** Unsure where an asset lives? The developer navigates to the LinkStream directory, types keywords into the global search bar, and finds the exact shortcut.

6. Alignment with Future Semester Milestones

- **Ticket 2 (Scrum Plan):** Roles (Product Owner, Scrum Master, Developers) mapped across link registry and redirect proxy epics.
- **Ticket 3 (DevOps / Code Management):** GitHub workflow with feature branches, squashing, merging, and automated CI test runs on PR creation.
- **Ticket 4 (Architecture Design):** Decomposed into 3 clean microservices:
 1. *Link Registry Service* (REST API CRUD)
 2. *Redirect & Cache Proxy* (Fast routing lookup)
 3. *Telemetry & Health Service* (Click counters and background URL pings).
- **Ticket 5 (Sprint Execution):** Full sprint execution with Docker containers, unit tests, and peer code reviews.